Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18BS5AI01]

FIFTH SEMESTER B.Tech EXAMINATION, MARCH - 2022

STATISTICS USING R

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Explain different methods of frequency distribution of statistical data and its constructions. **07**

b. Draw a pie diagram for the following data and analyze. Also explain how Rupee comes and goes. **07**

Rupee comes		Rupee goes	
	Paise		Paise
Excise	22	Central plan	25
Customs	18	Interest	15
Internal borrowing	18	Defence	13
Non-tax revenue	14	Share of taxes	14
Deficit	7	Other non-plan expenditure	12
Other Capital receipts	7	State & UT Plan assistance	12
Corporation tax	6	Subsidies	5
Income tax	3	Non-plan assistance	4
External assistance	3		
Other taxes	2		

2. a. Compute Mean deviation from median for the following data: **07**

Height in cms	158	159	160	161	162	163	164	165	166
No. of persons	15	20	32	35	33	22	20	10	8

Also compute coefficient of mean deviation.

b. Determine Karl Pearson's coefficient of skewness for the following frequency distribution. **07**

Variable	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	2	5	7	13	21	16	8	3

3. a. From the following data compute quantity indices by **07**

- **i.** Laspeyre' s method.
- **ii.** Paasche' s method.
- **iii.** Fisher' s method.

	2000		2002	
Commodity	Price	Total value	Price	Total value
A	10	100	12	180
B	12	240	15	450
C	15	225	17	340

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b. By analyzing the given data, obtain the weighted arithmetic mean index number and the weighted geometric mean index number. **07**

Commodity	Price in Rs./Unit		Weight
	Base year	Current year	
A	2	3	10
B	4	5	4
C	10	10	5
D	12	9	1

OR

4. a. Compute Fisher's index number from the following data. Show that it satisfies both time reversal and factor reversal tests and comments on the results. **07**

Item	Base year		Current year	
	Price	Quantity	Price	Quantity
A	6.00	10	6.40	12
B	4.50	20	5.50	24
C	2.30	15	4.00	16
D	6.10	6	8.00	10

b. Compute the chain index number with 2003 prices as base from the following table giving the average whole sale prices of commodities A, B, C for the year 2003 to 2007. **07**

Commodity	Average whole sale price (in Rs.)				
	2003	2004	2005	2006	2007
A	20	16	28	35	21
B	25	30	24	36	45
C	20	25	30	24	30

5. a. Calculate the trend value to the following data by the method of semi- averages. **07**

Year	1991	1992	1993	1994	1995	1996
Sales (in Rs.1000)	60	75	81	110	106	120

b. Determine the seasonal variations by simple average method for the data given below. **07**

Quarter

Year	I	II	III	IV
1989	30	40	36	34
1990	34	52	50	44
1991	40	58	54	48
1992	54	76	68	62
1993	80	92	86	82

OR

6. a. Discuss the mathematical models for a time series analysis. **05**

b. The following is the time series of turnover in crore rupees of a business firm. **09**

Year	1999	2000	2001	2002	2003	2004
Turnover (Crore upees)	6	9	11	7	7	10

Calculate the likely turnover in 2005 by fitting second-degree trend equation.

7. a. Explain the basic data types in R with examples. **04**

b. Write R command to create a matrix $A=\begin{bmatrix} 2 & 5 & 4 \\ 10 & 6 & 8 \\ 7 & 1 & 9 \end{bmatrix}$ and extract the following using R Commands **05**

- **i.** Elements at 2nd row,3rd column.
- **ii.** The entire 2nd row.
- **iii.** The entire 3rd column.
- **iv.** First and third column.

c. Write R commands to find **05**

- **i.** Relative frequency distribution of the eruption durations in faithful
- **ii.** The histogram of the eruption durations in faithful.

OR

8. a. Describe a vector in R and different operations on vectors with examples. **04**

b. Write R command to create a vector called V1 and V2 consisting of the numbers (2,3,4,5) and (2,4,6,8) respectively and perform the following using R commands **05**

- **i.** Add V1 and V2 store in a vector X1
- **ii.** Subtract V1 from V2
- **iii.** Create a new vector called "X2" by multiplying vector V1 by vector V2
- **iv.** Create a new vector called "X3" by taking the square root of each member of V1.

c. Write R commands to find **05**

- **i.** frequency distribution of the painter schools
- **ii.** bar graph of the painter schools and write a R command to color the bar chart.